

1. Let X and Y be sets, and let $\mathcal{A}$ and $\mathcal{B}$ be algebras of sets in X and Y . Let m be a premeasure on $\mathcal{A}$, and let n be a premeasure on $\mathcal{B}$. Show that if m and n are semifinite, then so is $m \otimes n$. Let $A \in \mathcal{A} \otimes \mathcal{B}$ with $m \otimes n(A) = \infty$. Let $r > 0$, and WLOG assume that $r > 1$. Note by definition of the product algebra there exists $A_1 \in \mathcal{A}, A_2 \in \mathcal{B}$ such that $A_1 \times A_2 = A$. The fact that $m \otimes n(A) = m(A_1)n(A_2) = \infty$ generates three cases
 - (a) if $m(A_1) = n(A_2) = \infty$, then by the semifinitude of n and m there exists $B_1 \subset A_1, B_2 \subset A_2$ such that $r < m(B_1), n(B_2) < \infty$, therefore $r < m(B_1)n(B_2) = m \otimes n(B_1 \times B_2) < \infty$, and $B_1 \times B_2 \in \mathcal{A} \otimes \mathcal{B}$.
 - (b) If $m(A_1) < \infty, n(A_2) = \infty$, then we can choose $B_2 \subset A_2$ such that $\frac{r}{m(A_1)} < m(B_2) < \infty$. Therefore $r = m(A_1)\frac{r}{m(A_1)} < m(A_1)n(B_2) = m \otimes n(A_1 \times B_2)$, and $A_1 \times B_2 \in \mathcal{A} \otimes \mathcal{B}$.
 - (c) if A_2 is finite and A_1 is infinite then the proof is the same as above.

Thus the product measure is semi-finite.

2. Let $X = Y = [0, 1]$, $\mathcal{M} = \mathcal{N}$ be the Borel σ algebra on $[0, 1]$, μ be the borel measure, ν be the counting measure on $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ respectively
 - (a) We claim that ν is not σ -finite. Let A_n be a sequence in $\mathcal{N}$ such that $\cup_{n=1}^{\infty} A_n = Y$. Since the union of the sets is countable, and Y is uncountable, implies there has to exists an uncountable set A_m . Thus the σ -finite condition of all sets in the sequence being finite is violated, as $\nu(A_m) = \infty$.
 - (b) Let $f : X \times Y \rightarrow \{0, 1\}$ be given by $f(x, y) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$. Note that to show that f is measurable we just need to compute $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$. Note that $f^{-1}(\{1\}) = \{(x, x) : x \in [0, 1]\}$. We can approximate this set via squares in the following way: consider the sequence $\{D_n\}_{n \in \mathbb{N}} \subset \mathcal{B}_{\mathbb{R}} \times \mathcal{B}_{\mathbb{R}}$ defined by $D_0 = [0, 1] \times [0, 1], D_n = \cup_{i=0}^{2^n-1} ([\frac{i}{2^n}, \frac{i+1}{2^n}], [\frac{i-1}{2^n}, \frac{i+1}{2^n}])$. Note that for an arbitrary point (x, y) where $x \neq y$, there exists an $m \in \mathbb{N}$ such that $\frac{1}{2^m} < |x - y|$, therefore $(x, y) \notin D_m$. Thus the countable intersection $\cap_{m=1}^{\infty} D_m \in \mathcal{B}_{\mathbb{R}} \times \mathcal{B}_{\mathbb{R}}$, therefore $f^{-1}(\{1\})$ is measurable. Since $f^{-1}(\{1\})^c = f^{-1}(\{0\})$, and the product σ -algebra is closed under complements, then $f^{-1}(\{0\})$ is measurable.
 - (c) We want to compute $\int_X (\int_Y f(x, \cdot) d\nu) d\mu$ and $\int_Y (\int_X f(\cdot, y) d\mu) d\nu$
 - For $\int_X (\int_Y f(x, \cdot) d\nu) d\mu$, note that $\int_Y f(x, \cdot) d\nu = 1$ since $f(x, \cdot) = 1_{\{x\}}(y)$ giving us $\int_Y f(x, \cdot) d\nu = \nu(\{x\}) = 1$. Therefore $\int_X (\int_Y f(x, \cdot) d\nu) d\mu = \int_X 1 d\mu = 1$
 - For $\int_Y (\int_X f(\cdot, y) d\mu) d\nu$, note that $\int_X f(\cdot, y) d\mu = 0$ by similar reasoning above, with the exception being that $\mu(\{y\}) = 0$. Thus the entire integral is 0.
3. Let $X = Y = \mathbb{N}$, and let $\mathcal{N} = \mathcal{M} = 2^{\mathbb{N}}$. Let $\nu = \mu$ be counting measures. Let $f(x, y) = \begin{cases} 1 & x = y \\ -1 & x = y + 1. \\ 0 & \text{otws} \end{cases}$. We must compute $\int_{X \times Y} |f(x, y)| d\mu \otimes \nu$, $\int_X (\int_Y f(x, \cdot) d\nu) d\mu$, and

$\int_Y (\int_X f(\cdot, y) d\mu) d\nu$. Note that f is measurable since $f^{-1}(1) = \{(x, x) : x \in \mathbb{N}\}$, $f^{-1}(-1) = \{(x + 1, x) : x \in \mathbb{N}\}$ are measurable, implying that $f^{-1}(0) = f^{-1}(\{1\}) \cup f^{-1}(\{-1\})$ is measurable. Note that

$$\begin{aligned} \int_{X \times Y} |f(x, y)| d\mu \otimes \nu &= \int_{f^{-1}(1)} |f(x, y)| d\mu \otimes \nu + \int_{f^{-1}(-1)} |f(x, y)| d\mu \otimes \nu \\ &= \sum_{n=1}^{\infty} \mu \otimes \nu(\{(n, n)\}) + \sum_{n=1}^{\infty} \mu \otimes \nu(\{(n+1, n)\}) \\ &= \sum_{n=1}^{\infty} \mu(\{n\})\nu(\{n\}) + \sum_{n=1}^{\infty} \mu(\{n+1\})\nu(\{n\}) \\ &= \sum_{n=1}^{\infty} 1 + \sum_{n=1}^{\infty} 1 \\ &= \infty \end{aligned}$$

$$\begin{aligned} \int_X (\int_Y f(x, \cdot) d\nu) d\mu &= f(1, 1)\nu(\{1\}) + \int_{X \setminus \{1\}} f(x, x)\nu(\{x\}) + f(x, x-1)\nu(\{x-1\}) d\mu \\ &= 1 + \sum_{x=2}^{\infty} (\nu(\{x\}) - \nu(\{x-1\}))\mu(x) \\ &= 1 \end{aligned}$$

$$\begin{aligned} \int_Y (\int_X f(\cdot, y) d\mu) d\nu &= \int_Y f(y, y)\mu(\{y\}) + f(y+1, y)\mu(\{y+1\}) d\nu \\ &= \sum_{y=1}^{\infty} (f(y, y) + f(y+1, y))\nu(\{y\}) \\ &= \sum_{y=1}^{\infty} (1 - 1) \\ &= 0 \end{aligned}$$

4. Let $f \in L^1(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ and let $g(x) = \sum_{n=-\infty}^{\infty} f(x+n)$. We want to show that $g(x)$ converges absolutely almost everywhere. Note that for each n we have that $f(x+n) = f \circ \tau_n$, where $\tau_n(x) = x+n$. It has been shown in theorem 5.25 that $\mu \circ \tau_n = \mu$.

Therefore, if we consider $\int_0^1 \sum_{n=-\infty}^{\infty} |f(x+n)| d\mu(x)$, we have that

$$\begin{aligned} \int_0^1 \sum_{n=-\infty}^{\infty} |f(x+n)| d\mu(x) &= \int_0^1 \sum_{n=-\infty}^{\infty} |f \circ \tau_n| d\mu \\ &= \sum_{n=-\infty}^{\infty} \int_0^1 |f \circ \tau_n| d\mu \\ &= \sum_{n=-\infty}^{\infty} \int_n^{n+1} |f| d\mu \\ &= \int_{\mathbb{R}} |f| d < \infty \end{aligned}$$

Therefore on the interval $[0, 1]$ g converges absolutely almost everywhere. Note that this same proof can be applied to an arbitrary interval $[m, m+1]$ for $m \in \mathbb{Z}$. Thus g converges absolutely for arbitrary $x \in \mathbb{R}$ almost everywhere.